

Researcher at the Group on Artificial Intelligence for People and Society (GAIPS, INESC-ID), working at the intersection of human-robot interaction, AI-driven assistive technologies, and physiological computing. With a background in Biomedical Engineering; Joana's current research applies cognitive modeling and signal processing to the design of autonomy-preserving cognitive support for neurodivergent children and older adults.

Website · **ORCID** · **Google Scholar**

Education

- 2025 **Unite! Winter Research School**, *University of Grenoble, France*
- 2023–Present **PhD in Computer Science and Engineering**, *Instituto Superior Técnico, Universidade de Lisboa, Portugal*
Supervised by Prof. Joana Campos (INESC-ID) and Dr. Filipa Correia (ITI/LARSyS).
- 2018–2021 **MSc in Biomedical Engineering**, *Technische Universität Wien, Austria*, Final average: 1.4/5 (excellent)
Major in Biomedical Signals & Instrumentation. Thesis: “*Elliptical method for the time-domain quantification of the arterial baroreflex*”, advised by Prof. Eugenijus Kaniusas and Univ. Ass. Babak Dabiri.
- 2015–2018 **BSc in Biomedical Engineering**, *Instituto Superior Técnico, Universidade de Lisboa, Portugal*, Final average: 16/20
- 2023 **Advanced Summer School on Artificial Intelligence (ESSAI & ACAI)**, *University of Ljubljana, Slovenia*

Academic & Research Experience

- 2023–Present **Researcher**, *Group of Artificial Intelligence for People and Society, INESC-ID, Lisbon, Portugal*
- 2023–Present **Teaching Assistant**, *Instituto Superior Técnico, Universidade de Lisboa, Portugal*
Artificial Intelligence; Logic for Programming.
- 2022–2023 **Research Assistant**, *Instituto de Telecomunicações, Lisbon, Portugal*
- 2019–2021 **Research Trainee**, *Technische Universität Wien, Austria*

Research Projects

- 2026–Present **PhD Student Fellow – FCT**, *Fundação para a Ciência e para a Tecnologia (2025.01710.BD)*
Design and evaluation of an assistive robot that provides support to individuals with cognitive impairments through autonomy-supportive interactions rather than task-oriented assistance alone.
- 2022–2025 **PhD Student Fellow – EMPOWER**, *European Commission (Horizon Europe, GA 101060918)*
Design and evaluation of technological support tools to empower stakeholders in digital education.
- 2022–2023 **Research Fellow – Neurosecurity and Experimental Communication**, *European Commission Joint Research Centre (JRC/IPR/2021/VLVP/3145)*

Publications

- 2026 **Brito, J.**; Belbute, M.; Stillwell, D.; et al. “Insights into Feature Design in a Multi-Game Framework for Executive Function Support in Neurodivergent Children.” *International Journal of Human-Computer Studies* Submitted.
- 2026 **Brito, J.**; Neerincx, A.; Neerincx, A.; Soares, A.; et al. “Workshop YOUR Study Design 2025!” *Proceedings of the 2026 ACM/IEEE International Conference on Human-Robot Interaction*, Edinburgh. doi:10.1145/3757279.3785542

- 2026 Mohan, M.; **Brito, J.**; et al. "Workshop YOUR Study Design 2026!" *Companion of the 2026 ACM/IEEE International Conference on Human-Robot Interaction*, Edinburgh. doi:10.1145/3776734.3793852
- 2025 **Brito, J.**; Brito Duarte, R.; Fonseca, H. C. "Assistant Robots with an Agenda Foster Uncooperative Behaviors." *2025 IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, Eindhoven. doi:10.1109/RO-MAN63969.2025.11217891
- 2024 Mohan, M.; **Brito, J.**; Neerincx, A.; Alves-Oliveira, P.; Fitter, N. T. "Workshop YOUR Study Design 2024!" *ACM/IEEE HRI*, Boulder, CO. doi:10.1145/3610978.3638167
- 2024 Neerincx, A.; **Brito, J.**; Couto, M.; Campos, J.; et al. "Social Robotics in Psychological Interventions for Children." In *Digital Technologies for Learning and Psychological Interventions*. Springer Nature. doi:10.1007/978-3-031-76414-1_6
- 2024 Bota, P.; **Brito, J.**; Fred, A.; Cesar, P.; Silva, H. "A Real-World Dataset of Group Emotion Experiences Based on Physiological Data." *Scientific Data*. doi:10.1038/s41597-023-02905-6
- 2021 Dabiri, B.; **Brito, J.**; Kaniusas, E. "Cardiovagal Baroreflex Hysteresis Using Ellipses in Response to Postural Changes." *Frontiers in Neuroscience* 15:720031. doi:10.3389/fnins.2021.720031

Service & Activities

- 2026 Student volunteer, ACM/IEEE International Conference on Human-Robot Interaction (HRI'26, CORE A*), Edinburgh.
- 2025 Proceedings Chair, Social Media Chair & Student Volunteer Chair, IEEE 2025 Conference on Games (CoG'25, CORE B), Lisbon.
- 2024–Present Organising Committee, Workshop "YOUR Study Design", ACM/IEEE International Conference on Human-Robot Interaction (HRI'24 in Boulder, CO, HRI'25 in Melbourne, HRI'26 in Edinburgh).
- 2024–Present Social Media Team of the Group of Artificial Intelligence for People & Society, INESC-ID.
- 2024 Student volunteer, 4th TAILOR Conference "Trustworthy AI from Lab to Market", Lisbon.

Awards

- 2021 Academic Distinction, Technische Universität Wien (awarded to students with average below 1.5).
- 2018 Diploma of Academic Merit, Instituto Superior Técnico (top 10% of students, 2015–2018).

Research Interests



Human-Robot Interaction: Child-Robot Interaction, Social Robotics, Assistive Technologies



Digital Health & Serious Games: Neurodevelopmental Disorders, Cognitive Assessment, AI-driven Tools



Physiological Computing: Biosignals, Affective Computing, Wearable Sensors

Languages

Portuguese native

English C1